

1. A

2. A

3. C

4. A

5. A

1.

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

3S 3C 4R 4H 6M 1C 7H 1N 4W 6S 5Y 5N

Take a look at the following data. Here, the Y label indicates whether a child goes out to play or not.

1. (12 points, 3 points for each attribute) Calculate the information gain for each attribute and select the optimal splitting attribute (to construct the first layer of the tree).
2. (8 points) Draw the final tree structure without writing out the calculation process. (Hint: The error rate of the final training set is 0, that is, all samples are correctly classified.)

(1). $Info(S|A) = \sum_{v=1}^{V} \frac{|D_v|}{|S|} \times Info(D_v)$ $Gain(A) = Info(S) - Info(S|A)$

$Info(S) = -\frac{1}{2}\log(\frac{1}{2}) - \frac{1}{2}\log\frac{1}{2} = 1$

① weather : $Info(S|A) = \frac{3}{10}(-\frac{1}{3}\log\frac{1}{3} - \frac{2}{3}\log\frac{2}{3}) + \frac{3}{10}(-1\log 1) + \frac{4}{10}(-\frac{1}{4}\log\frac{1}{4} - \frac{3}{4}\log\frac{3}{4})$

$Gain(A) \approx 1 - 0.5998 \approx 0.4002$

② Temperature : $Info(S|A) = \frac{4}{10}(-\frac{1}{2}\log\frac{1}{2} - \frac{1}{2}\log\frac{1}{2}) + \frac{5}{10}(-\frac{3}{5}\log\frac{3}{5} - \frac{2}{5}\log\frac{2}{5}) + \frac{1}{10}(-1\log 1)$

$Gain(A) \approx 1 - 0.8855 = 0.1145$

③ Humidity : $Info(S|A) = \frac{7}{10}(-\frac{3}{7}\log\frac{3}{7} - \frac{4}{7}\log\frac{4}{7}) + \frac{3}{10}(-\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3})$

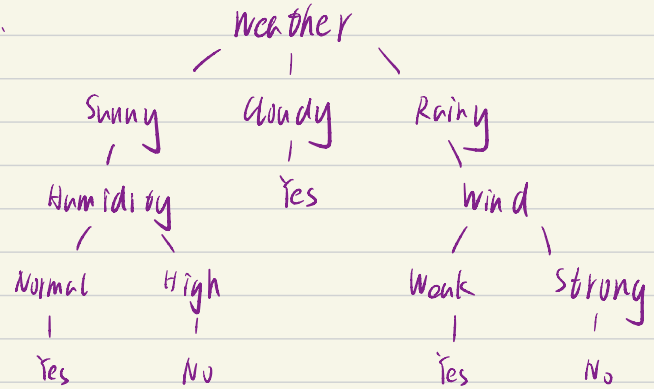
$Gain(A) \approx 1 - 0.9649 \approx 0.0351$

④ Wind : $Info(S|A) = \frac{7}{10}(-\frac{3}{4}\log\frac{3}{4} - \frac{1}{4}\log\frac{1}{4}) + \frac{6}{10}(-\frac{2}{3}\log\frac{2}{3} - \frac{1}{3}\log\frac{1}{3})$

$Gain(A) \approx 1 - 0.8752 \approx 0.1248$

So weather has the highest information gain.

(2).



2.

1.2.(CNN, 20 points) Consider the convolutional network defined by the layers below. The input shape is $28 \times 28 \times 1$ and the output is 5 neurons. Consider layers:

$$\text{Conv4(15)} + \text{Maxpool}_2 + \text{Conv2(25)} + \text{Maxpool}_2 + \text{FC5}$$

where

- Conv4(15): 15 filters with each size $4 \times 4 \times D$, where D is the depth of the activation volume at the previous layer, stride $= 1$, padding $= 1$;
- Conv2(25): 25 filters with each size $2 \times 2 \times D$, where D is the depth of the activation volume at the previous layer, stride $= 1$, padding $= 0$;
- Maxpool_2: 2×2 filter, stride $= 2$, padding $= 0$;
- FC5: A fully-connected layer with 5 output neurons.

1. (10 pts) Compute the shape of activation map of each layer.
2. (10 pts) Compute the total number of parameters of each layer.

(1). $(N + 2p - F) / \text{stride} + 1$

Conv 4(15): $(28 + 2 - 4) / 1 + 1 = 27$

$27 \times 27 \times 15$ shape

$(4 \times 4 \times 1 + 1) \times 15 = 255$ parameters

Maxpool_2: $(27 - 2) / 2 + 1 = 13$

$13 \times 13 \times 15$ shape

No parameter

Conv 2(25): $(13 + 0 - 2) / 1 + 1 = 12$

$12 \times 12 \times 25$ shape

$(2 \times 2 \times 15 + 1) \times 25 = 1525$ parameters

Maxpool_2: $(12 - 2) / 2 + 1 = 6$

$6 \times 6 \times 25$ shape

No parameter

FC5 :

5×1 shape

$(6 \cdot 6 \cdot 25) \times 5 + 5 = 4505$ parameters